

NLP-III

Q1. What is the IDF of a term that occurs in every document? Compare this with the use of stop word lists.

Let term be t and documents be n ($d_1, d_2, d_3, \dots, d_n$)

According to question, t appears in each d , meaning total documents containing term is also n i.e $df(t) = n$

$$\text{Then IDF} = \log \frac{\text{Number of documents}}{\text{Total documents containing term}} = \log \frac{n}{n} = \log 1 = 0$$

- *An IDF of 0 means the term has zero discriminative power or informativeness across the collection.*
- *Multiplying this zero weight by the term frequency effectively nullifies the word's impact in Tf-idf weighting scoring*

Q1. What is the idf of a term that occurs in every document? Compare this with the use of stop word lists.

IDF vs Stop words:

Functionally equivalent - same outcome for ubiquitous words -> ignored or stripped of influence during retrieval or scoring

Mechanism	Stop Words <i>explicitly remove common words (like "the", "and", "is") from the vocabulary entirely before processing. This reduces index size and saves memory.</i>	IDF weighting <i>Keeps the words in the vocabulary and index but automatically assigns them a weight of 0 (or near-zero), mathematically neutralizing their impact without requiring a predefined exclusion list.</i>
Flexibility	<i>Stop word lists require manual curation and might miss or incorrectly remove context-dependent terms.</i>	<i>IDF handles domain-specific high-frequency noise automatically based on the specific corpus.</i>

Q2. Consider the following tables of term frequencies and document frequencies (df), and document collection sizes (N). Calculate for each term-document-pair the tf-idf weight with $\text{idf} = \ln(N/\text{df})$, use raw term frequencies. Show all intermediate steps.

term	doc 1	doc 2	doc 3	df	N
car	27	4	24	18,136	94,584
auto	3	33	0	6,723	53,814
insurance	0	33	29	19,241	97,226
best	14	0	17	25,235	113,095

		TF-IDF		
term	IDF	doc 1	doc 2	doc 3
car	1.651	44.593		
auto	2.08			
insurance				
best				

		TF-IDF		
term	IDF	doc 1	doc 2	doc 3
car	1.65	44.55	6.6	39.6
auto	2.08	6.24	68.64	0
insurance	1.62	0	53.46	46.98
best	1.5	21	0	25.5

Q3. What is the minimum and maximum value of the idf that a term can have (assuming a document frequency larger than zero)? In which cases does this happen?

The minimum value of IDF is 0, and the maximum value depends on the total number of documents (N) in the corpus.

i.e.

Max value = $\log(N)$ -> occurs if term occurs in single document

Min value – 0 -> if term occurs in every document

Maximum value of idf_t for a term t is $\log(N)$ (term occurs in single document).

Minimum value of idf_t for a term t is 0 (term occurs in every document).

Q4. Why the idf of a term always finite?

- DIY

Q5. Consider the corpus with three documents:

D1: the cats are in the house

D2: the dogs are in the house and outside

D3: the cats and dogs are friends

Assume no preprocessing (normalization, case folding, stop word removal) is done, determine the important terms from the corpus using TF-IDF weighing. Represent these documents with TF-IDF

Term	TF			IDF	TF-IDF		
	D1	D2	D3		D1	D2	D3
the	$2/6 = 1/3$	$2/8 = 1/4$	$1/6$	$\ln(3/3) = 0$	0	0	0
cats	$1/6$	0	$1/6$	$\ln(3/2) = 0.405$	0.0675	0	0.0675
are	$1/6$	$1/8$	$1/6$	$\ln(3/3) = 0$	0	0	0
in	$1/6$	$1/8$	0	$\ln(3/2) = 0.405$	0.0675	0.0506	0
house	$1/6$	$1/8$	0	$\ln(3/2) = 0.405$	0.0675	0.0506	0
dogs	0	$1/8$	$1/6$	$\ln(3/2) = 0.405$	0	0.0506	0.0675
and	0	$1/8$	$1/6$	$\ln(3/2) = 0.405$	0	0.0506	0.0675
outside	0	$1/8$	0	$\ln(3/1) = 1.09$	0	0.136	0
friends	0	0	$1/6$	$\ln(3/1) = 1.09$	0	0	0.136

Term	TF			IDF	TF-IDF		
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cats	$1/6$	0	$1/6$	$\ln(3/2) = 0.405$	0.0675	0	0.0675
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in	$1/6$	$1/8$	0	$\ln(3/2) = 0.405$	0.0675	0.0506	0
house	$1/6$	$1/8$	0	$\ln(3/2) = 0.405$	0.0675	0.0506	0
dogs	0	$1/8$	$1/6$	$\ln(3/2) = 0.405$	0	0.0506	0.0675
and	0	$1/8$	$1/6$	$\ln(3/2) = 0.405$	0	0.0506	0.0675
outside	0	$1/8$	0	$\ln(3/1) = 1.09$	0	0.136	0
friends	0	0	$1/6$	$\ln(3/1) = 1.09$	0	0	0.136

TF-IDF Representation

[the cats are in house dogs and outside friends]

[[[**0** 0.0675 0 0.0675 0.0675 0 0 0 0]
[**0** 0 0.0506 0.0506 0.0506 0.0506 0.136 0]
[**0.0675** 0 0 0 0.0675 0.0675 0 0.136]]]

Q6. Represent the following three sentences using TF-IDF. Please remove stop words. No stemming and Lemmatization are applied. Use natural log. Stop words list: {I, we, it, are, be, is, am, the, will}.

sentence 1: The weather forecast says it will be sunny tomorrow.

sentence 2: I like sunny weather.

sentence 3: The assignment is due tomorrow.

- After preprocessing (stop word removal only), sentences become:
- sentence 1: weather forecast says sunny tomorrow
sentence 2: like sunny weather
sentence 3: assignment due tomorrow
- Solution- DIY

Term	TF			IDF	TF-IDF		
	S1	S2	S3		S1	S2	S3
weather							
forecast							
says							
sunny							
tomorrow							
like							
assignment							
due							